

Article

The Impact of ENSO Shocks on Firm Performance: The Role of Supply Chain Resilience and Network Complexity in Energy Firms

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Abstract

Escalating climate volatility, particularly the El Niño/Southern Oscillation (ENSO), poses severe operational and financial risks to corporate sustainability in the energy sector. However, quantitative evidence regarding how macro-level climate shocks transmit to micro-level operational performance remains scarce. Integrating dynamic capability and social network theories, this study analyzes a panel of 103 Chinese listed energy firms (2005–2022) using System GMM, mediation, and moderation models. The results indicate that ENSO intensity significantly impairs performance; specifically, a 1 °C rise in sea surface temperature anomalies decreases firms' return on assets (ROAs) by 0.142%. We identify supply chain resilience as a critical strategic mechanism for climate adaptation, where response capacity acts as the dominant mediating channel, while recovery capacity functions as an independent compensatory mechanism. Conversely, supply network complexity—across horizontal, vertical, and spatial dimensions—amplifies the negative impact of climate disruptions by hindering resource mobility. Heterogeneity analysis reveals that state-owned enterprises exhibit stronger institutional resilience, and firms in southern regions partially offset impacts through hydropower advantages. This study bridges climate science with operations management, offering strategic guidance for managers to configure resilient, sustainable supply chains capable of withstanding environmental turbulence.

Keywords: supply chain resilience; supply network complexity; firm performance; ENSO; energy firms; climate adaptation



Academic Editors: Mou Leong Tan,
Cheng Li, Fei Zhang and Kwok
Pan Chun

Received: 3 February 2026

Revised: 18 March 2026

Accepted: 19 March 2026

Published: 26 March 2026

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1. Introduction

ENSO is a primary factor driving interannual climate variability, characterized by its quasi-periodic nature and extensive global teleconnections [1], with extreme phases known as El Niño and La Niña. ENSO significantly alters the frequency and intensity of extreme weather events [2], adversely disrupting agricultural production [3], commodity markets [4], and overall social stability [5]. With climate change, the intensity and frequency of ENSO events are expected to increase [6,7], potentially leading to exponential economic losses [8]. Projections suggest that the resulting losses could soar to \$84 trillion by the close of the 21st century [9], highlighting the growing socioeconomic risks associated with these changes.

Energy systems underpin modern economies [10], and the firms operating them are central to climate adaptation. While ENSO's impact on energy supply and demand is established [11], how these climate shocks precisely propagate to impair firm performance is unclear. Grounded in dynamic capability theory, extreme events from ENSO (e.g., droughts, floods) impair infrastructure and hydropower efficiency [12], oil transportation [13], and critical mineral supplies [14]. Such physical shocks often exceed firms' dynamic capabilities to buffer and reconfigure resources and undermine energy firms' higher-order dynamic capabilities—the strategic capacities to sense, reconfigure, and learn that are essential for adapting to environmental change. Combined with social network theory, supply network structures further propagate localized shocks [15], as seen in the cross-border price volatility from regional droughts [16]. The 2015–2016 El Niño exemplified such cascading failures, inflicting systemic economic losses far surpassing direct damages [8]. Thus, this study proposes first question: does ENSO negatively affect the performance of energy firms?

Dynamic capabilities are influenced by both internal and external factors [17]. Internally, supply chain resilience involves the ability to anticipate, respond to, and recover from disruptions [18], aligning with the processes of sensing, seizing, and reconfiguring processes essential for the dynamic capability framework [19]. For instance, absorptive capacity helps anticipate disruptions through strategies like improved hydrological forecasting [20]; response capacity enables the reallocation of resources during disturbances, such as fostering bioenergy collaborations to stabilize operations; and recovery capacity addresses long-term restoration, particularly in energy infrastructure, where asset specificity often delays recovery. These considerations lead to the second research question: does supply chain resilience mediate the ENSO-energy firms' performance relationship?

Externally, social network theory offers insights into how the structural characteristics of supply networks influence the spread of shocks [21]. Climate-related disruptions are often intensified by multiple dimensions of complexity, which horizontal ties facilitate the spread of disruptions, vertical layers contribute to information asymmetry, and spatial breadth increases exposure to region-specific risks. A case in point is the 2015 El Niño drought in Indonesia, which not only impaired agricultural output but also hindered port and transport operations, causing cascading effects across coal, palm oil, and related sectors [22]. A third question is thus posed: does supply network complexity moderate ENSO's impact on the performance of energy firms?

Our study aims to address these compelling yet unanswered questions. This study develops an empirical framework grounded in dynamic capabilities and social network theories, examining the pathways and boundary conditions through which ENSO disruptions influence firm performance via internal resilience (supply chain resilience) and external structural characteristics (supply network complexity). The results reveal a significant negative direct influence of ENSO on firm performance, where a 1 °C rise in SST anomalies corresponds to a 0.142% decline in return on assets. Supply chain resilience mediates this relationship, predominantly through response capacity, marginally through absorption, and independently through recovery, reflecting decoupled dynamic capabilities under asset specificity and delayed recovery in energy sectors. Moreover, supply network complexity significantly amplifies ENSO shocks via heightened interdependencies and coordination failures.

This study offers key contributions. First, it shifts the focus from established macro-level ENSO effects—such as energy price fluctuations [23], market linkages [24], and inflationary trends [25]—to underexplored micro-level transmission mechanisms within supply chains. By applying a hierarchical capability lens and the sensing-seizing-reconfiguring framework, we present the first micro-evidence from the energy sector on how climate

disruptions propagate internally, systematically revealing cross-scale pathways and empirically informing adaptation strategies.

Second, recent research is increasingly examining supply chain resilience aspects like operational flexibility or digitalization [26], but most treat it as monolithic or isolate its dimensions [27,28], which limits understanding of its dynamic nature. This study employs social network analysis to delineate three distinct capacities (absorption, response, and recovery) based on structural and relational features of corporate supply networks. This approach quantifies resilience mechanisms by accounting for interdependencies and resource flows, offering a dynamic perspective on how energy firms endure climate volatility.

Last, while supply network complexity is often associated with elevated coordination costs and disruption risks [29], it may also foster adaptation through diversity and redundancy under high social capital or IT integration [30,31]. Given mixed empirical findings on its moderating effect [32,33], particularly regarding climate shocks, this study clarifies complexity's role as an amplifier of ENSO disruptions, offering empirical clarity to the “is it too complex?” debate and supporting climate-aware network design.

The remainder of the paper is organized as follows. Section 2 reviews the literature and hypotheses. Section 3 outlines the data and methodology, followed by empirical results in Section 4. Section 5 discusses implications and conclusions.

2. Literature Review and Hypothesis Development

2.1. ENSO and Energy Firms' Performance

The extant literature indicates that ENSO is a key climate risk factor driving volatility in global energy markets [34,35]. This volatility leads to fluctuations in energy stock prices and instability in the industry by damaging infrastructure and disrupting production and manufacturing logistics (see Table 1). While macro-level evidence shows market fluctuations, the micro-level supply chain transmissions through which ENSO affects energy firms' performance remain a theoretical “black box”, hindering targeted resilience strategies.

Table 1. Key research on the impact of ENSO on energy systems.

Research Focus	Author (Year)	Key Research Objective	Relevant Findings
Energy Market Volatility and Pricing	Cashin et al. (2017) [14]	To analyze El Niño's global macroeconomic transmission	El Niño drives short-term inflation via higher energy prices, with heterogeneous growth impacts
	Nam (2021) [36]	To assess ENSO's effect on global commodities	El Niño induces inflationary pressures across energy and non-energy commodities
	Hong et al. (2023) [23]	To forecast crude oil volatility using the Southern Oscillation Index (SOI) trend	The SOI trend component boosts crude oil volatility forecast accuracy
Renewable Energy Investment and Financial Markets	Wei et al. (2022) [37]	To explore ENSO's heterogeneous effects on renewable stocks	Asymmetric impacts on EU renewable stocks
	Wei et al. (2023) [38]	To quantify connectedness among ENSO, carbon, oil, and renewable stocks	ENSO is a net short-term information contributor; renewables and oil are key spillover bridges
	Zhang et al. (2024) [39]	To evaluate ENSO-fossil energy market linkages under extreme conditions	ENSO acts as a net spillover transmitter during market extremes
Macroeconomic and Policy	Zhang et al. (2023) [40]	To compare effects of physical (ENSO) vs. text-based climate indicators on Japan's EPU	ENSO shocks increase economic policy uncertainty, unlike text-based indicators
Hydropower Supply	Trespalcios et al. (2023) [41]	To analyze El Niño's impact on electricity prices in a hydro-dependent system	El Niño-induced scarcity increases price mean, variance, and tail risk

Dynamic capability theory offers a valuable lens for unpacking this “black box”, elucidating how firms maintain competitiveness through resource transformation amid changing conditions [42]. This framework posits a hierarchical organization of capabilities, differentiating higher-order dynamic capabilities from operational routines [43]. These

superior capabilities—such as sensing, coordinating, reconfiguring, and learning—enable firms to systematically adapt their resource bases and internal processes. We argue that ENSO's negative influence arises mainly from the erosion of these higher-order capabilities [44], which represent strategic capacities to adjust to environmental shifts [43], and that this erosion occurs through three interconnected pathways.

Operationally, ENSO-induced extreme weather events, such as droughts reducing hydropower output [45], directly damage physical assets and disrupt production. These direct impacts test a firm's ability to reconfigure operational routines and tangible resources, a key dynamic capability in crises [19]. Structurally, resource access and risk depend on node criticality and relational topology [15]. Disruptions at critical nodes (e.g., mining regions or transmission corridors) propagate rapidly due to high connectivity, triggering cascading effects like inventory shortages and import reliance [46]. Such structural disruptions distort information, weaken threat sensing, and hamper coordinated action, extending losses beyond initial physical impacts, as observed during the 1997–1998 and 2015–2016 El Niño episodes [8]. In market terms, ENSO exacerbates commodity price volatility and shifts demand [47]. For example, lowering heating needs and thus consumption of natural gas and heating oil [48]. It also elevates sector-wide systemic risk by intensifying climate risk perception [34], which in turn limits strategic flexibility and obstructs organizational learning.

In summary, ENSO undercuts firm performance by degrading essential higher-order dynamic capabilities, namely reconfiguration, sensing, coordination, and learning. We therefore hypothesize:

H1: *ENSO negatively affects the performance of energy firms.*

2.2. Mediating Role of Supply Chain Resilience

Supply chain resilience denotes a system's ability to anticipate, absorb, and rebound from disruptions [49]. Recent work highlights how technology and structural reconfiguration can improve responsiveness [50], yet resilience is frequently treated as a unitary construct. This obscures the distinct capacities involved, particularly in asset-intensive energy contexts.

To clarify these mechanisms, we adopt a dynamic capability perspective, specifically the sensing-seizing-reconfiguring framework [19], which offers a process-oriented view of adaptation. These dimensions align with established resilience capacities (absorptive, response, and recovery capacities), reflecting adaptive, absorptive, and innovative capabilities at the supply chain level [51,52].

We conceptualize three sequential resilience dimensions accordingly. Absorptive capacity, central to the sensing phase, enables firms to mitigate initial disruptions by converting climate signals into preventive measures, such as “climate loans” that transform meteorological data into financial instruments. Response capacity, aligned with seizing, allows dynamic resource reallocation and process adjustment during disruptions—for instance, using predictable El Niño fire emission patterns to pre-position firefighting assets or reroute logistics [53]. Recovery capacity functions in the reconfiguration phase, involving the restoration of normal operations. In energy sectors, however, recovery is often delayed by high asset specificity, as seen in the extended lead times for custom transmission components [54], though external supports (e.g., policy interventions) may ease such constraints. We thus hypothesize:

H2a: *Energy firms' exposure to ENSO reduces firm performance by impairing absorptive capability within their supply chains.*

H2b: *Energy firms' exposure to ENSO reduces firm performance by impairing response capability within their supply chains.*

H2c: *Energy firms' exposure to ENSO reduces firm performance by impairing recovery capability within their supply chains.*

2.3. Moderating Role of Supply Network Complexity

Supply network complexity is a critical structural determinant of resilience, yet its role in recurring climate oscillations like ENSO remains understudied. While complexity can buffer localized shocks, it also propagates systemic disruptions due to interdependencies [55]. In climate contexts, high complexity may elevate operational emissions [56] and weaken governance, eroding relational benefits [30]. However, extant work focuses on static complexity or isolated disruptions (e.g., Toyota's 2010 recall; COVID-19), overlooking dynamic, recurrent ENSO effects—particularly salient for energy firms with spatially fixed and multi-tiered infrastructures.

To address this, we draw on social network theory, which posits that a firm's structural position governs resource access and coordination under uncertainty [57]. Network ties facilitate coordination under uncertainty [58], and network architecture shapes information and material flows, crucial for resilience to systemic disruptions like ENSO [46]. Supply network complexity, commonly captured via horizontal, vertical, and spatial dimensions [59], reflects actor multiplicity and interdependence [60], raising coordination costs and structural inertia [61]. Thus, social network theory complements dynamic capabilities by specifying structural constraints on resilience in networked environments.

We argue that complexity moderates ENSO shock transmission by altering network-mediated flows. Horizontal complexity increases nodal interdependence, amplifying contagion as follows: dense, centralized networks accelerate disruption spread (e.g., single-supplier flooding paralyzing multiple generators), while partner heterogeneity impedes coordinated adaptation [62]. Strong ties may hasten domino effects; weak ties, though useful for novel resources, often degrade under systemic strain [63]. Vertical complexity introduces inertia and information distortion across tiers, fostering bullwhip effects as climate uncertainties trigger hoarding and inflation [64]. The primary friction here is administrative and informational: deep tier layers create severe information asymmetry and rigid contract structures, which structural bottlenecks and institutional misalignment [61] further constrain reconfiguration. Spatial complexity extends networks geographically, heightening vulnerability to ENSO's regional variability. Hydropower shortfalls in one region boost global gas demand [65], while dispersed nodes face divergent recovery due to regulatory and infrastructural heterogeneity [66]. Geographic dispersion essentially creates profound friction by exposing supply chains to physical logistics vulnerabilities, such as distant port or rail disruptions. Furthermore, this structural spread significantly amplifies the administrative burden, making it exceedingly difficult to coordinate recovery efforts across multiple regional jurisdictions that often operate under uncoordinated or conflicting climate emergency protocols.

In summary, supply network complexity amplifies ENSO disruptions by restricting resource mobility, distorting information, and increasing synchronization failures. We hypothesize:

H3a: *Horizontal complexity negatively moderates ENSO's impact on energy firms' performance.*

H3b: *Vertical complexity negatively moderates ENSO's impact on energy firms' performance.*

H3c: *Spatial complexity negatively moderates ENSO's impact on energy firms' performance.*

The research framework of this study is shown in Figure 1.

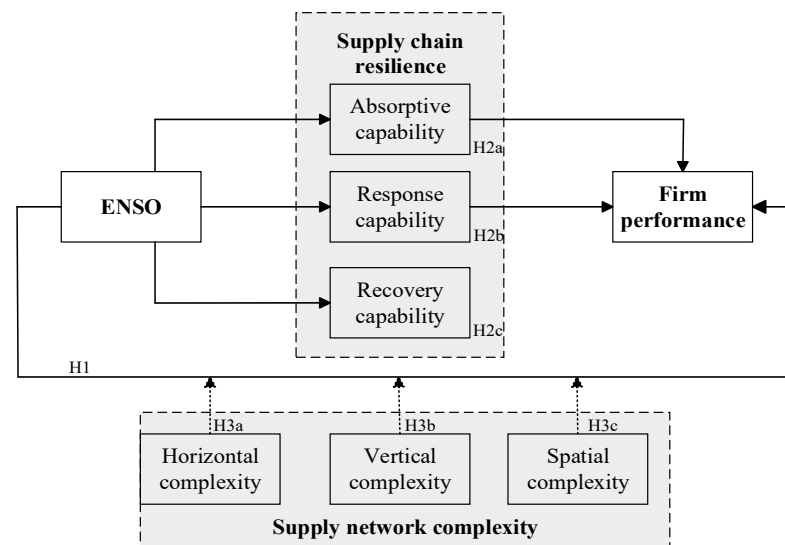


Figure 1. Research framework.

3. Data and Methodology

3.1. Research Setting and Data Collection

This study focuses on publicly listed firms within China's energy sector, which is highly climate-sensitive, making it an ideal context for exploring climate-supply chain connections. As the world's top energy consumer (27.6% of global primary energy in 2023) and carbon emitter (31% of global CO₂ in 2022), China plays a decisive role in energy transition and climate governance. Moreover, concentrating on a single major economy also helps control for cross-country institutional differences while preserving variation across energy subsectors, thus facilitating clearer insight into resilience mechanisms. The sample encompasses key energy industries, as summarized in Table 2.

Table 2. Major enterprises in the sample.

Industry Code	Industry Name	Firms (N)	Proportion (%)
D44	Electricity and heat production/supply	49	47.57%
B06	Coal mining and washing	16	15.53%
B11	Mining support activities	10	9.71%
C25	Petroleum, coking, and nuclear fuel processing	10	9.71%
D45	Gas production and supply	9	8.74%
Others	Other industries (including Water Production and Supply, Oil and Gas Extraction, and Wholesale Trade)	9	8.74%

A unique panel dataset was constructed from 2005 to 2022, spanning multiple ENSO cycles (including the 2015–2016 super El Niño). Firm-level data came from the China Stock Market and Accounting Research Database (CSMAR) and Chinese Research Data Services (CNRDS) platform, supplemented with sector-level from the National Bureau of Statistics and the China Energy Statistical Yearbook. ENSO was measured using annual sea surface temperature (SST) anomalies in the El Niño 3.4 region, as provided by the U.S. National Oceanic and Atmospheric Administration (NOAA).

The sample was refined by excluding ST (Special Treatment)/ST (Risk of Delisting)/PT (Particular Transfer) firms, entities listed for under one year, those delisted or suspended, and records with incomplete key variables. The final dataset consists of 1751 firm-year observations across 103 energy companies. To minimize the influence of outliers, continuous variables were winsorized at the 2nd and 98th percentiles.

3.2. Dependent Variable

This study employs return on assets (ROAs) to evaluate firm performance, defined as net profit divided by total assets. ROA directly captures asset utilization efficiency and is less affected by market fluctuations [67,68]. Alternative metrics such as Tobin's Q can be influenced by market sentiment and industry cycles [69], while the P/E ratio is sensitive to accounting policies and short-term earnings distortions [70]. Given the asset-intensive structure of the energy sector, ROA more directly reflects the impact of physical climate shocks on asset productivity, rendering it particularly suitable for assessing ENSO-related disruptions.

3.3. Core Independent Variable

ENSO intensity is typically measured via sea surface temperature (SST) anomalies or the Southern Oscillation Index (SOI). Following common practice [24,71], we use the Niño 3.4 index (5° N–5° S, 170° W–120° W), in which a blend of regions 3 and 4 is known for reliably capturing ENSO dynamics. Monthly SST deviations are computed relative to a 30-year baseline. El Niño events are identified when anomalies exceed +1 standard deviation and remain above +0.5 °C for at least eight consecutive months; La Niña episodes are defined symmetrically. Thus, persistent positive SST anomalies in the Niño 3.4 region (proxied by NINO34) serve as the direct measure of El Niño.

3.4. Mediating Variables

Drawing on prior frameworks [72,73], we conceptualize supply chain resilience as three sequential capacities—absorptive (ASC), response (RSC), and recovery (RCC). These are quantified using social network analysis [74] to capture structural features of the supply chain. We first identify the top five suppliers and customers of each firm from database records to build node-level networks. Python 3.10-based web crawlers retrieve corresponding stock data to ensure accuracy. We then construct dynamic, yearly annual undirected affiliation networks to track temporal shifts in supply relationships and analyze in Gephi 0.10 to derive specific metrics. Indicator definitions are summarized in Table 3. Finally, to quantify the three composite dimensions, we employed the Entropy Weight Method (EWM). This method objectively assigns weights to each sub-indicator based on its informational entropy and data dispersion, aggregating them into the final ASC, RSC, and RCC indices used as explanatory variables. The detailed formulas for all sub-indicators are provided in Table A1.

Table 3. Supply chain resilience indicator system.

Constructs	Sub-Indicators	Conceptual Description	Adapted From
<i>Absorptive capability (ASC)</i>	Customer concentration	The firm's degree of reliance on a few key downstream clients	Pham & Nguyen (2024) [75]
	Supplier concentration	The firm's degree of centralization on specific upstream supply sources	Wang et al. (2026) [76]
	Supply chain emphasis	Management's attention to supply chain transparency via annual report textual analysis	Jüttner & Maklan (2011) [77];
	Closed-loop supply chain level	The firm's interconnectedness within the overall supply network (Degree Centrality)	Freeman (1978) [78]

Table 3. Cont.

Constructs	Sub-Indicators	Conceptual Description	Adapted From
<i>Response capability (RSC)</i>	Supply demand matching	The extent of demand information distortion across the supply chain (bullwhip effect)	Yang et al. (2020) [79]
	Customer stability	The firm's ability to sustain long-term, reliable downstream sales channels	Ou (2024) [80]
	Supplier stability	The consistency of reliable upstream partnerships to buffer supply risks	Shishodia et al. (2022) [81]; Chen et al. (2025) [82]
<i>Recovery capability (RCC)</i>	Supply chain network community structure	The density of internal connectivity within localized network sub-communities (Modularity)	Namdar et al. (2024) [83]; Mijbas et al. (2025) [84]
	Supply chain network location	The firm's integration level based on adjacent nodes' influence (Eigenvector Centrality)	Raj et al. (2022) [85]; Li et al. (2023) [86]
	Supply chain network status	The structural significance of the firm within the global network (PageRank)	Wang et al. (2023) [87]

Note: detailed formulas and data sources for all indicators are provided in Table A1.

3.5. Moderators

Horizontal Complexity (HC). Existing measures, such as basic supplier counts [64] or the Herfindahl–Hirschman Index (HHI) [88], have significant limitations. Basic supplier counts ignore product portfolio heterogeneity—meaning firms with the same number of suppliers might face vastly different coordination challenges if their product diversity differs [89]. Meanwhile, HHI relies on highly granular procurement volume data that are often inaccessible for multi-product firms. To better reflect coordination intensity across product lines, we measure HC as the number of first-tier suppliers scaled by a weighted count of distinct product categories [89], accounting for variations in portfolio breadth and depth (as shown in Equation (1)).

$$HC_{i,t} = \frac{N_{i,t}}{1 / \sum_{k=1}^K p_{i,t,k}^2} \quad (1)$$

where N represents the breadth (total number of distinct product segments), and $\rho_{i,k}^2$ captures the depth (revenue concentration).

Vertical Complexity (VC). VC denotes the average number of second-tier suppliers per first-tier supplier, capturing multi-tier dependency depth [56]. Traditional approaches (e.g., tier counts, longest hierarchical path) [90,91] fail to quantify cross-tier interaction density, overlooking how similar tier structures may entail divergent risks due to varying dispersion among lower-tier suppliers [92]. Higher VC amplifies the probability of minor disruptions triggering domino effects. This paper evaluates the spread of secondary suppliers to better represent risk propagation and addresses the limitations of hierarchical methods.

Spatial Complexity (SC). This study quantifies SC using a procurement-weighted geodesic distance model [92], formalized as

$$SC = \sum_{j=1}^4 \left(\frac{P_{ij} \times D_{ij}}{D_{i,Total}} \right) \quad (2)$$

where P_{ij} is the proportion of firm i 's procurement volume in region j (Europe, North America, Asia/Pacific, rest of the world), D_{ij} is the great-circle distance between the firm's headquarters (at latitude lat_i , longitude lon_i) and the geographic centroid of region j , $D_{i,Total}$ and represents the sum of the geodesic distances across all regions for firm i , computed via

$$D_{ij} = \frac{2\pi r}{360} \arccos[\cos(lat_i)\cos(lat_j)\cos(lon_i - lon_j) + \sin(lat_i)\sin(lat_j)] \quad (3)$$

with Earth's radius $r = 6378$ km. By integrating volume-proportional weighting, physical distance friction, and centroid-based normalization, our approach quantifies true logisti-

cal burden more accurately than count-based indices [93] or foreign-supplier proportion methods [94], while enabling cross-firm comparability.

3.6. Control Variables

To ensure robust identification and mitigate confounding effects, we include firm/industry controls informed by the prior literature [35,95–97]. Firm-level controls include: Climate Vulnerability (CLVA), measured by operating profit to net fixed assets, indicating physical risk exposure; Size (log of total assets) and Age (log of years operating) to address scale and lifecycle effects; Capital Intensity (CAI) (capital expenditure to revenue), reflecting asset-heavy operations; Leverage (LEV) (long-term debt to market value), addressing financial risk; and Slack Resources (SSRs) (administrative expenses to revenue), signaling adaptive capacity. Industry controls encompass the Energy Consumption Structure (ECP) (coal share in total energy) and Energy Fixed Capital (FC) to capture energy dependency and asset commitments. Time dummies for the 2007–2009 financial crisis (CRISIS) and 2020–2022 pandemic (PANDE) absorb major macro shocks.

3.7. Model Specification

3.7.1. System Generalized Method of Moments (SYS-GMM)

We incorporate a lagged dependent variable to capture performance persistence and dynamic adjustment in energy firms, which may introduce correlation with the error term due to unobserved firm heterogeneity and potential endogeneity. Conventional estimators such as OLS and fixed effects are inconsistent under these conditions [98]. We therefore employ System GMM [99], a method well-suited to dynamic panels with large N and small T. This estimator combines difference and level equations, instrumenting differences with lagged levels and vice versa, improving efficiency over difference-GMM by retaining initial observations and exploiting additional orthogonality conditions [99]. Model validity is evaluated via the Arellano–Bond test for autocorrelation and Hansen test for instrument overidentification [100]. The model is specified as Model (4).

$$ROA_{it} = \beta_0 + \beta_1 ROA_{it-1} + \beta_2 NINO34_{it} + \beta_3 CLVA_{it} + \beta_4 Size_{it} + \beta_5 Age_{it} + \beta_6 CAI_{it} + \beta_7 LEV_{it} + \beta_8 SSR_{it} + \beta_9 ECP_{it} + \beta_{10} FC_{it} + \beta_{11} CRISIS_{it} + \beta_{12} PANDE_{it} + \epsilon_{it} \quad (4)$$

where ROA_{it} denotes firm i 's performance in year t , $ROA_{i,t-1}$ is its lag. $NINO34_{it}$ reflects the ENSO intensity, while other variables represent control variables, and ϵ_{it} is the error term.

3.7.2. Mediating Effect Model

To examine whether supply chain resilience mediates the ENSO–performance link, we follow the causal steps approach [101]. Model (5) estimates the impact of ENSO (β_2 for NINO34) on mediators M (absorptive, response, and recovery capabilities). Model (6) then assesses the direct effect of ENSO on ROA (β_2') and the mediator's influence (β_3). Full mediation occurs when β_2 in Model (5) is significant, β_3 is significant but β_2' is not in Model (6); partial mediation occurs when all three variables are significant and $|\beta_2'| < |\beta_1|$ (where β_1 is ENSO's total effect in Model (4)), with the mediation proportion calculated as $(\beta_2\beta_3)/\beta_1$. If the stepwise coefficients (β_2 or β_3) are insignificant, we utilize Bootstrap testing with 5000 resamples to directly evaluate the confidence intervals of the indirect effect to confirm mediation robustness:

$$M_{it} = \beta_0 + \beta_1 M_{it-1} + \beta_2 NINO34_{it} + \beta_3 CLVA_{it} + \beta_4 Size_{it} + \beta_5 Age_{it} + \beta_6 CAI_{it} + \beta_7 LEV_{it} + \beta_8 SSR_{it} + \beta_9 ECP_{it} + \beta_{10} FC_{it} + \beta_{11} CRISIS_{it} + \beta_{12} PANDE_{it} + \epsilon_{it} \quad (5)$$

$$ROA_{it} = \beta_0 + \beta_1 ROA_{it-1} + \beta_2' NINO34_{it} + \beta_3 M_{it} + \beta_4 CLVA_{it} + \beta_5 Size_{it} + \beta_6 Age_{it} + \beta_7 CAI_{it} + \beta_8 LEV_{it} + \beta_9 SSR_{it} + \beta_{10} ECP_{it} + \beta_{11} FC_{it} + \beta_{12} CRISIS_{it} + \beta_{13} PANDE_{it} + \epsilon_{it} \quad (6)$$

3.7.3. Moderating Effect Model

Beyond these direct effects, two-way interactions indicate the synergy between variable pairs [64]; therefore, we further examine whether supply network complexity—horizontal (HC), vertical (VC), and spatial (SC)—moderates the ENSO–ROA relationship through Models (7) to (9):

$$ROA_{it} = \beta_0 + \beta_1 ROA_{it-1} + \beta_2 NINO34_{it} + \beta_3 HC_{it} + \beta_4 (NINO34_{it} \times HC_{it}) + \beta_5 CLVA_{it} + \beta_6 Size_{it} + \beta_7 Age_{it} + \beta_8 CAI_{it} + \beta_9 LEV_{it} + \beta_{10} SSR_{it} + \beta_{11} ECP_{it} + \beta_{12} FC_{it} + \beta_{13} CRISIS_{it} + \beta_{14} PANDE_{it} + \epsilon_{it} \quad (7)$$

$$ROA_{it} = \beta_0 + \beta_1 ROA_{it-1} + \beta_2 NINO34_{it} + \beta_3 VC_{it} + \beta_4 (NINO34_{it} \times VC_{it}) + \beta_5 CLVA_{it} + \beta_6 Size_{it} + \beta_7 Age_{it} + \beta_8 CAI_{it} + \beta_9 LEV_{it} + \beta_{10} SSR_{it} + \beta_{11} ECP_{it} + \beta_{12} FC_{it} + \beta_{13} CRISIS_{it} + \beta_{14} PANDE_{it} + \epsilon_{it} \quad (8)$$

$$ROA_{it} = \beta_0 + \beta_1 ROA_{it-1} + \beta_2 NINO34_{it} + \beta_3 SC_{it} + \beta_4 (NINO34_{it} \times SC_{it}) + \beta_5 CLVA_{it} + \beta_6 Size_{it} + \beta_7 Age_{it} + \beta_8 CAI_{it} + \beta_9 LEV_{it} + \beta_{10} SSR_{it} + \beta_{11} ECP_{it} + \beta_{12} FC_{it} + \beta_{13} CRISIS_{it} + \beta_{14} PANDE_{it} + \epsilon_{it} \quad (9)$$

4. Analyses and Results

Descriptive statistics and correlations are presented in Table A2. The strong correlation between ROA and its lagged term ($\beta = 0.52$) supports the dynamic specification. ENSO is significantly negatively correlated with ROA, offering preliminary support for H1.

4.1. Direct Effect Analysis

Table 4 shows that ENSO significantly reduces firm performance ($\beta = -0.142$; $p < 0.1$). A 1 °C increase in the Niño3.4 SST anomaly reduces ROA by 0.142%, consistent with recent views [102,103] that environmental factors, particularly major climatic oscillations like ENSO, critically shape energy-sensitive industries' economic performance. Moreover, the significantly positive coefficient on lagged ROA confirms performance persistence.

Table 4. Direct effects of ENSO on energy firms' performance and the moderation of supply network complexity.

Model	(1) Main Effects		(2) Two-Way Interactions	
	1	2	3	4
Main effects				
L.ROA	0.212 *** (0.050)	0.198 *** (0.044)	0.217 *** (0.044)	0.213 *** (0.057)
NINO34	−0.142 * (0.081)	−0.136 * (0.077)	−0.134 * (0.077)	−0.133 * (0.080)
Horizontal		0.179 (0.180)		
Vertical			0.444 (0.279)	
Spatial				−0.045 * (0.024)
Two-way interaction effects				
NINO34 × Horizontal		−0.316 * (0.182)		
NINO34 × Vertical			−0.499 ** (0.254)	
NINO34 × Spatial				−0.024 * (0.015)

Table 4. Cont.

Model	(1) Main Effects	(2) Two-Way Interactions		
	1	2	3	4
<i>Main effects</i>				
<i>Controls</i>				
CLVA	7.675 *** (2.947)	11.92 *** (1.999)	11.53 *** (2.019)	5.300 (3.510)
Size	−0.706 (0.491)	−0.083 (0.263)	−0.099 (0.272)	−0.018 (0.153)
Age	−0.015 (0.038)	−0.025 (0.040)	−0.019 (0.043)	−0.020 (0.0294)
CAI	−0.647 (0.937)	−0.990 (0.724)	−1.224 (0.762)	−0.579 (0.537)
LEV	−0.192 *** (0.039)	−0.170 *** (0.036)	−0.167 *** (0.035)	−0.411 *** (0.131)
SSR	−27.09 *** (7.833)	−10.14 (6.845)	−12.58 * (6.943)	−19.64 ** (8.327)
ECP	3.799 (3.838)	2.913 (2.605)	2.613 (2.916)	7.079 *** (2.430)
FC	0.402 (0.254)	0.118 (0.227)	0.126 (0.232)	0.532 ** (0.242)
CRISIS	0.084 (0.334)	0.591 ** (0.296)	0.643 * (0.350)	0.194 (0.343)
PANDE	−0.126 (0.405)	−0.355 (0.246)	−0.364 (0.260)	−0.115 (0.340)
Constant	0.920 (25.85)	−8.297 (15.53)	−6.605 (16.86)	−29.10 ** (11.91)
Observations	1751	1751	1751	1751
Controls	YES	YES	YES	YES
Firms ^a	103	103	103	103
Instrument variables	34	99	77	30
Hansen test ^b	0.151	0.333	0.217	0.391
AR(1)	0	0	0	0
AR(2) ^c	0.470	0.509	0.542	0.827

Notes. Standard errors in parentheses. Endogenous variables: ROA, CLVA, Age; instruments: lags t-2 to t-3. Exogenous: ECP, FC, CRISIS, all interactions. Predetermined variables use lags t-1 to t-2. Collapsed instruments limit instrumental variable counts to sections while preserving validity. ^a 2005–2022. ^b *p*-value. null: instruments exogenous. ^c *p*-value. null: no second-order autocorrelation. * *p* < 0.1, ** *p* < 0.05, *** *p* < 0.01.

4.2. Mediating Effect Analysis

According to Table 5, ENSO significantly weakens absorptive capacity (ASC) in Model 1 ($\beta = -0.144$; $p < 0.1$), implying that a 1 °C SST rise reduces ASC by 0.144%. Model 2 further reveals that ASC positively affects ROA ($\beta = 0.019$; $p < 0.05$), while the direct effect of ENSO on ROA remains significantly negative ($\beta = -0.128$; $p < 0.1$) even after accounting for the mediator, indicating partial mediation. ASC accounts for 1.927% of the total effect, supporting H2a and the pathway of “ENSO—reduced absorptive capacity—lower performance”.

Model 3 indicates ENSO impairs response capability (RSC) ($\beta = -4.302$; $p < 0.05$). Model 4 shows both ENSO ($\beta = -0.183$; $p < 0.05$) and RSC ($\beta = 0.017$; $p < 0.1$) significantly influence ROA. Notably, the direct effect of ENSO intensifies from -0.142 to -0.183 after including RSC, suggesting a masking effect. When response capability is not activated, the adverse effects of ENSO on performance are exacerbated. RSC mediates 51.50% of the total effect, confirming RSC as a key transmission channel and establishing hypothesis 2b.

Model 5 indicates that the model does not validate the impact of ENSO on recovery capacity (RCC), suggesting that ENSO does not directly impair RCC. To rigorously address the inherent trend-stationarity of energy infrastructure networks, we explicitly incorporate a deterministic linear time trend (specified as a continuous variable, Trend = 1, 2, ..., 18, corresponding to the 2005–2022 panel) in the RCC models. Unlike agile operational capabilities (ASC and RSC), recovery capability is rooted in physical network topology characterized by prolonged investment cycles and immense sunk costs, causing it to evolve

with a long-term structural drift. By controlling for this time trend, the autoregressive coefficient of RCC (L.RCC) is estimated at 0.891. This methodological refinement corrects potential unit-root biases in the dynamic panel and statistically confirms the extreme structural inertia of energy supply networks. Model 6 reveals that RCC positively influences ROA ($\beta = 0.117$; $p < 0.1$) while ENSO remains negative ($\beta = -0.001$; $p < 0.1$). This indicates that RCC serves a compensatory regulatory function, operating independently of ENSO while still contributing slightly to the improvement of ROA. As suggested by the unique characteristics of the energy sector, this decoupling is largely driven by a “structural lag”. Given the high asset specificity, immense sunk costs, and prolonged investment cycles inherent to energy infrastructure (e.g., transmission networks and power plants), physical recovery is structurally rigid. Consequently, recovery capabilities are far less sensitive to immediate, year-to-year climate fluctuations compared to agile operational response mechanisms. Although hypothesis H2c is not supported, the findings underscore the dynamic complementarity of resilience capabilities.

Table 5. Intermediary role test of the supply chain resilience.

Model	1 ASC	2 ROA	3 RSC	4 ROA	5 RCC	6 ROA
L.ROA		0.243 *** (0.057)		0.127 ** (0.051)		0.199 *** (0.046)
L.ASC	0.371 *** (0.100)					
ASC		0.019 ** (0.010)				
L.RSC			0.274 *** (0.062)			
RSC				0.017 * (0.010)		
L.RCC					0.891 *** (0.00)	
RCC						0.117 * (0.154)
NINO34	−0.144 * (0.084)	−0.128 * (0.077)	−4.302 ** (1.947)	−0.183 ** (0.088)	−0.00 * (0.001)	−0.001 * (0.001)
Constant	67.05 (63.39)	9.087 (23.06)	285.2 *** (68.26)	−0.041 (0.199)	−0.084 * (0.187)	−0.674 (0.309)
Observations	1751	1751	1751	1751	1751	1751
Controls	YES	YES	YES	YES	YES	YES
Firms ^a	103	103	103	103	103	103
Instrument variables	24	45	24	25	35	38
Hansen test ^b	0.289	0.134	0.106	0.383	0	0.356
AR(1)	0	0	0	0	0	0
AR(2) ^c	0.433	0.584	0.929	0.226	0.539	0.661

Notes. Standard errors in parentheses. Endogenous variables in models 1, 3, 5: ASC, RSC, RCC, CLVA, Age; instruments: lags t-2 to t-3. Endogenous in models 2, 4, 6: ROA, CLVA, Size, Age, Cash; instruments: lags t-2 to t-4. Exogenous: ECP, FC, SSR. Predetermined variables use lags t-1 to t-2 (models 1, 3, 5) or t-1 to t-3 (others). Collapsed instruments limit instrumental variable counts to sections while preserving validity. ^a 2005–2022. ^b p -value. null: instruments exogenous. ^c p -value. null: no second-order autocorrelation. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3. Moderation Analysis

As shown in Table 4, all three complexity dimensions exacerbate ENSO’s adverse effect on performance. Horizontal complexity intensifies the ENSO-ROA link in Model 2 ($\beta = -0.316$; $p < 0.1$), as do vertical in Models 3 ($\beta = -0.499$; $p < 0.05$) and spatial complexity ($\beta = -0.024$; $p < 0.1$) in Models 4. These indicate that greater linkage, hierarchical, or geographical complexity magnifies El Niño-induced profitability loss.

To further clarify the moderating effect of supply chain complexity, we plot ENSO’s marginal effects on ROA in Figure 2. Following Aiken and West (1991) [104], trajectories reflect high/low complexity (mean $\pm$ 1 SD) across dimensions. Results show that high

horizontal (HC) and vertical complexity (VC) steepen ENSO's negative slope, accelerating firm performance decline. In contrast, spatial complexity (SC) slightly weakens ENSO's marginal effect, which remains negative and significant.

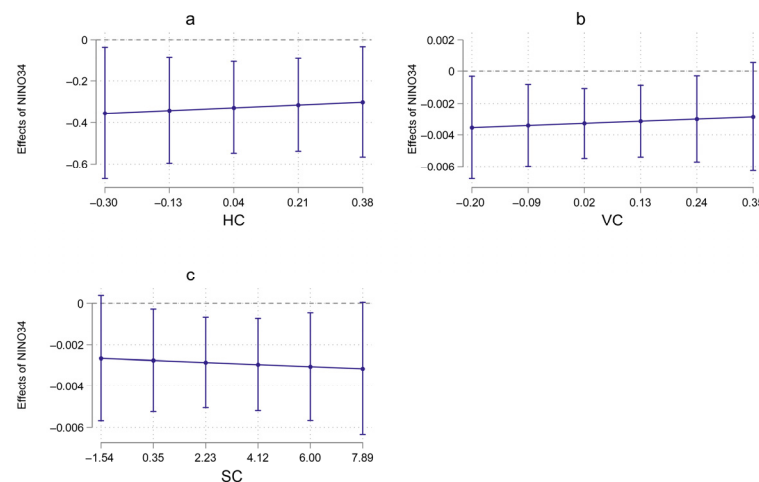


Figure 2. Moderation effects: (a) Moderating effect of horizontal complexity (HC); (b) Moderating effect of vertical complexity (VC); (c) Moderating effect of spatial complexity (SC).

4.4. Robustness Checks

A series of robustness checks were conducted to validate the core findings, with consistent results reported in Table A3 (Models 1–5).

Dependent Variable Alternative. Return on equity (ROE) is used to measure firm performance as it specifically reflects returns on shareholders' equity and is more responsive to capital structure than ROA [68]. Results are presented in Model 1 in Table A3.

Independent variable alternative. ENSO is replaced by the Pacific Decadal Oscillation (PDO), which captures longer-term Pacific climate variability and is often phase-locked with El Niño. Warmer PDO phases typically coincide with El Niño [105], making the PDO useful for verifying climate-economic mechanisms and long-term effects. Results are presented in Model 2 in Table A3.

Adjusting sample window. We restrict the sample [106] to moderate-to-strong ENSO years (2006–2007, 2009–2010, and 2014–2016) to account for intensity threshold effects, as weaker events tend to have limited climatic impacts. This approach highlights economically significant climate shocks. Results are presented in Model 3 in Table A3.

Propensity score matching (PSM). To mitigate self-selection bias, samples are split at the median NINO34 value [107], with the above-median group as treated (treatment = 1) and the rest as control (treatment = 0). Propensity scores are estimated via logistic regression and matched using nearest-neighbor matching (caliper = 0.001). Results are presented in Model 4 in Table A3.

Nonlinear expansion. We include a squared NINO34 term (Nino) to test for potential nonlinear effects [108], ensuring that results are not sensitive to the assumed linear functional form. Results are presented in Model 5 in Table A3.

Placebo Test. A randomization placebo test is conducted to address unobservable factors [109]. We randomly reassign NINO34 values to generate placebo treatments, performing 500 permutations with random numbers to ensure exogeneity from actual ENSO shocks. The resulting coefficients are normally distributed around zero (see Figure 3), confirming that unobserved factors do not drive our results.

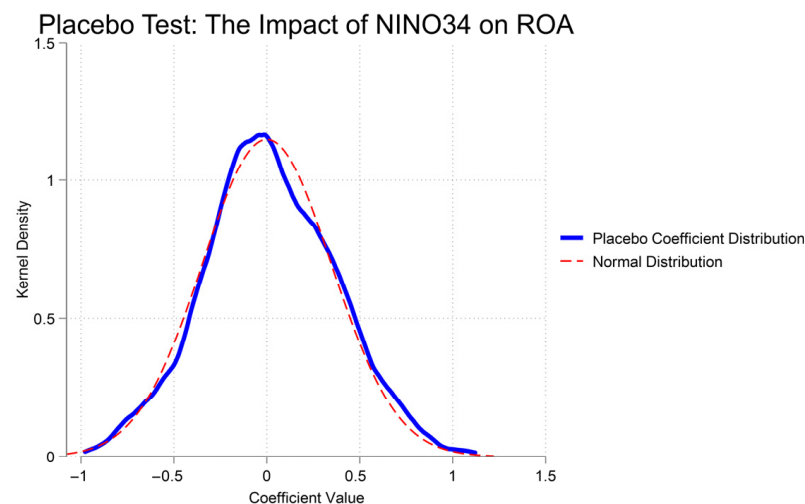


Figure 3. Placebo test.

4.5. Further Analysis

Regional Climate Sensitivity. ENSO spatially amplifies precipitation over southern China, with highly climate-sensitive areas identified within 105° E–120° E and 18° N–27.5° N (Guangdong, Guangxi, Hainan, Fujian, Hunan, and southern Jiangxi) and the remainder classified as low sensitivity [110]. Table A3 shows ENSO improves performance for energy firms in high-sensitivity regions (Model 6: $\beta = 0.6$; $p < 0.05$) but reduces it in low-sensitivity areas (Model 7: $\beta = -0.219$; $p < 0.01$). Increased rainfall enhances hydropower generation in southern provinces [111], particularly in mountainous reservoir clusters areas like Guangxi and Fujian. Meanwhile, northern inland areas face water shortages that strain thermal power operations and coal supply chains [102].

Corporate Ownership Heterogeneity. Firms exhibit distinct climate risk exposures based on ownership structure. When grouping firms into state-owned (SOEs) and non-state-owned enterprises (NSOEs), Table A3 reveals no significant ENSO effect on SOEs (Model 8), but a significant negative impact on NSOEs (Model 9: $\beta = -0.319$; $p < 0.1$). SOEs benefit from government support and invest in resilient infrastructure that mitigate physical supply chain vulnerabilities, while NSOEs suffer due to capital constraints and market-driven exposure to climate shocks [112].

5. Conclusions and Implications

Integrating dynamic capability and social network theories, this study moves beyond macroeconomic correlations to establish a micro-founded understanding of how ENSO climate shocks impair energy firm performance. The energy sector is characterized by high asset specificity, critical infrastructure dependencies, and an inherent sensitivity to climatic conditions, making it a paramount yet underexplored domain for climate impact research. Using a dynamic panel (2005–2022) and SYS-GMM to address endogeneity, including reverse causality and unobserved heterogeneity, we establish causal pathways linking climate shocks to firm outcomes. Empirically validated with diagnostic tests, we then found that this effect is differentially mediated by resilience capacities and amplified by network complexity, while regional and ownership structures shape heterogeneous responses. These findings advance climate resilience frameworks and offer practical insights for energy firms and policymakers. Key conclusions include:

- (1) ENSO directly impairs energy firm performance by constraining resource reconfiguration capacities and amplifying systemic network risks. A 1 °C SST anomaly increase lowers ROA by 0.142%. This negative impact remains robust across multiple ro-

bustness checks, confirming ENSO as a critical external disruptor of operational and financial stability in resource-intensive sectors.

- (2) Supply chain resilience as a differentiated mediator in responding to ENSO disruptions. Absorptive capacity acts as a minor mediator (1.927% of total effect), serving as an initial buffer, while response capacity constitutes the dominant mediating channel (51.503%), underscoring that the core vulnerability lies in the inability to dynamically reconfigure resources and coordinate responses after a shock occurs. In contrast, recovery capacity does not function as a mediator but operates independently, exerting a compensatory positive influence on ROA, which aligns with its role as a long-term restorative mechanism often reliant on external support rather than immediate buffering.
- (3) Supply network complexity (horizontal, vertical, and spatial) significantly drove losses by increasing structural interdependencies and coordination failures. Marginal effects indicate that high horizontal and vertical complexity steepens performance decline, whereas spatial complexity remains detrimental despite slight attenuation at high levels.
- (4) There is notable heterogeneity among regions and firm types regarding climate impacts. In high-sensitivity regions like South China, firms benefit from increased hydropower output, challenging the uniform “victim” narrative, while low-sensitivity regions suffer significant losses. State-owned enterprises are largely insulated from impacts due to government support, whereas non-state-owned enterprises face severe negative effects, highlighting substantial institutional and adaptive disparities.

These findings carry significant implications for both corporate managers and policy-makers. At the firm level, first, energy companies should prioritize climate resilience as a core strategic objective rather than a peripheral concern. Crucially, for non-state-owned enterprises and firms lacking natural geographic hedges such as hydropower advantages, managers must proactively build synthetic resilience. To achieve this, companies can utilize financial instruments like weather derivatives or climate insurance to actively hedge against operational disruptions. Second, diversifying energy procurement portfolios is essential to mitigate single-region reliance, while establishing cross-regional capacity-sharing alliances allows firms to dynamically route energy resources when local grids face ENSO-induced stress. Finally, integrating such redundancy planning into multi-tiered and geographically dispersed networks will further ensure robust operational continuity during extreme environmental turbulence.

At the policy level, first, prioritize climate-resilient infrastructure investment in low-sensitivity regions with limited natural buffers and high ENSO vulnerability. Second, provide ownership-tailored policy support (e.g., targeted liquidity tools, energy storage subsidies), focusing on boosting pre-shock preparedness and real-time response capabilities of non-state-owned enterprises. Finally, promote regulatory frameworks that enhance transparency, standardization, and information sharing across complex energy supply networks to curb systemic risk propagation.

This study offers empirical evidence on ENSO’s impacts on energy firms but has limitations requiring future exploration. First, it focuses solely on the El Niño phase; incorporating La Niña would enable comparative analysis of both phases’ asymmetrical climatic effects. Second, generalizability may be limited to the energy sector; extending the analysis to other climate-sensitive industries (e.g., agriculture, logistics) using primary data would strengthen external validity. Third, a critical boundary condition of this study is its primary focus on the physical risks of climate change. In reality, energy firms also face simultaneous transition risks driven by stringent external policy environments. Future research should explore the complex interaction between physical climate extremes and

long-term decarbonization mandates to provide a more holistic understanding of the sustainability trade-offs facing energy systems.

Author Contributions: Conceptualization, X.L.; Methodology, K.G.; Software, A.L.; Validation, X.L. and Y.Y.; Investigation, K.G. and A.L.; Data curation, X.D.; Writing—original draft, X.L.; Writing—review & editing, K.G.; Supervision, X.D. and Y.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research is supported by the 2024 “Revelation and Leadership” Project Foundation of Chongqing Jiaotong University; the Ministry of Education Humanities and Social Sciences Fund, grant number 24YJA630023; the Heilongjiang Provincial Philosophy and Social Science Research Planning Project, grant number 23GLC040; the Heilongjiang Higher Education Teaching Reform Project, grant number SJGYB2024516, and the Heilongjiang Provincial University Basic Research Business Fund, grant number 130212122039.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

Table A1. Supplementary Notes on the indicators of supply chain resilience.

Constructs/Indicators	Mathematical Formula	Data Source
Absorptive Capability (ASC)		
Customer concentration (CC)	$CC_{it} = \frac{\text{Sales to Top 5 Customers}_{it}}{\text{Total Annual Sales}_{it}}$	CSMAR
Supplier concentration (SC)	$SC_{it} = \frac{\text{Purchases from Top 5 Suppliers}_{it}}{\text{Total Annual Purchases}_{it}}$	CSMAR
Supply chain emphasis (SCE)	$SCE_{it} = \ln(1 + \text{Sentence count of supply chain keywords in MD\&A}_{it})$ Measured via Degree Centrality in an undirected affiliation network.	CSMAR/Textual Analysis
Closed-loop supply chain level (CL)	Evaluated as the total number of unique supplier and customer nodes directly connected to firm.	Derived via Gephi
Response Capability (RSC)		
Supply demand matching (Matching_{it})	$\text{Matching}_{it} = \frac{\frac{\sigma(\text{Production sup})_{it}}{\mu(\text{Demand sup})_{it}}}{\frac{\sigma(\text{Production cus})_{it}}{\mu(\text{Demand cus})_{it}}}$ σ : Standard deviation of the volume. μ : Mean volume. sup (supply): Upstream production metrics, proxied by sales volume. cus (customer): Downstream demand metrics, proxied by the cost of goods sold.	CSMAR/CNRDS
Customer stability (CST)	$CST_{it} = \frac{\text{Number of identical Top 5 customers in year } t \text{ and } t-1}{5}$	CSMAR
Supplier stability (SST)	$SST_{it} = \frac{\text{Number of identical Top 5 suppliers in year } t \text{ and } t-1}{5}$	CSMAR
Recovery Capability (RCC)		
Supply chain network community structure	Evaluated via modularity. Quantifies the density deviation of connections within localized network communities versus random networks.	Derived via Gephi
Supply chain network location	Evaluated via Eigenvector Centrality. Measures a node's influence weighted by the centrality of its connected partners.	Derived via Gephi
Supply chain network status	Evaluated via PageRank Centrality. A link analysis algorithm assessing global structural importance within the network topology.	Derived via Gephi

Table A2. Descriptive statistics and correlations.

	ROA	LROA	NINO34	ASC	RSC	RCC	HC	VC	SC	CLVA	Size	Age	CAI	LEV	SSR	ECP	FC	CRISIS	PANDE
ROA	1																		
LROA	0.52 ***	1																	
NINO34	−0.07 ***	−0.01	1																
ASC	−0.02	−0.04	−0.09 ***	1															
RSC	0.01	0.04	0.04	0.18 ***	1														
RCC	−0.04	−0.04	0	0.14 ***	0.16 ***	1													
HC	−0.02	−0.03	0.03	0.04	0.11 ***	0.15 ***	1												
VC	−0.01	−0.02	0.03	0.04	0.10 ***	0.13 ***	0.93 ***	1											
SC	−0.07 ***	−0.06 **	0.11 ***	0.19 ***	0.43 ***	0.35 ***	0.39 ***	0.36 ***	1										
CLVA	0.74 ***	0.45 ***	−0.07 ***	0.01	−0.01	−0.04	−0.01	−0.01	−0.04	1									
Size	0.09 ***	0.08 ***	0.01	0.10 ***	0.10 ***	−0.02	0.15 ***	0.16 ***	0.07 ***	0.10 ***	1								
Age	−0.15 ***	−0.17 ***	−0.02	0.23 ***	0.25 ***	−0.01	0.10 ***	0.10 ***	0.30 ***	−0.12 ***	0.22 ***	1							
CAI	0.01	0.08 ***	0.01	−0.04	0.01	0.01	0.08 ***	0.09 ***	−0.02	0.03	0.12 ***	−0.08 ***	1						
LEV	−0.39 ***	−0.31 ***	−0.01	−0.01	0	0.13 ***	0	0.01	0.02	−0.32 ***	0.04	0.14 ***	0	1					
SSR	−0.14 ***	−0.10 ***	0.04	−0.11 ***	−0.07 ***	−0.07 ***	−0.08 ***	−0.08 ***	−0.11 ***	−0.18 ***	−0.43 ***	−0.15 ***	0.09 ***	−0.11 ***	1				
ECP	0.14 ***	0.15 ***	0.03	−0.31 ***	−0.25 ***	0	−0.11 ***	−0.10 ***	−0.41 ***	0.06 **	−0.30 ***	−0.61 ***	0.11 ***	−0.01	0.12 ***	1			
FC	−0.11 ***	−0.08 ***	0.10 ***	0.13 ***	0.17 ***	0	0.15 ***	0.15 ***	0.39 ***	−0.06 **	0.25 ***	0.43 ***	−0.08 ***	0.06 ***	−0.07 ***	−0.67 ***	1		
CRISIS	0.10 ***	0.09 ***	−0.03	−0.10 ***	−0.14 ***	0	−0.11 ***	−0.11 ***	−0.28 ***	0.03	−0.16 ***	−0.30 ***	0.09 ***	−0.03	0.48 ***	−0.48 ***	0.05 **	1	
PANDE	−0.07 ***	−0.08 ***	−0.32 ***	0.35 ***	0.12 ***	0	0.05 **	0.04	0.16 ***	−0.02	0.18 ***	0.43 ***	−0.07 ***	−0.01	−0.67 ***	0.25 ***	−0.09 ***	−0.20 ***	1
MEAN	0.028	0.028	−0.151	0.100	0.241	0.280	0.149	0.085	3.179	0.093	22.915	13.404	0.195	4.399	0.075	4.173	10.011	0.167	0.167
SD	0.055	0.055	1.155	0.175	0.112	0.145	0.556	0.323	4.719	0.240	1.478	6.532	0.200	5.437	0.064	0.095	0.507	0.373	0.373

Notes. Standard errors in parentheses. ** $p < 0.05$, *** $p < 0.01$.

Table A3. Robustness checks and heterogeneity analysis.

Model	1 ROE	2 ROA	3 ROA	4 ROA	5 ROA	6 High-Sensitive ROA	7 Low-Sensitive ROA	8 State-Owned ROA	9 Non-State-Owned ROA
NINO34	−0.019 * (0.010)		−0.238 ** (0.120)			0.600 ** (0.271)	−0.219 *** (0.079)	−0.084 (0.076)	−0.319 * (0.188)
LROE	−0.034 *** (0.009)								
LROA		0.131 *** (0.038)	0.267 ** (0.114)		0.241 *** (0.054)	0.116 ** (0.059)	0.197 *** (0.050)	0.090 *** (0.029)	0.236 ** (0.098)
PDO		−0.605 * (0.354)							
Treatment				−0.852 ** (0.338)					
Nino					−0.114 * (0.068)				
Constant	−6.439 (4.755)	−7.224 (18.05)	−4.665 (51.44)	−35.77 * (19.68)	−27.12 ** (11.95)	−116.8 *** (35.55)	−15.02 (36.18)	2.434 (24.82)	−96.14 * (54.81)
Observations	1751	1751	721	596	1751	310	1441	1261	490
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
Firms ^a	103	103	103	103	103	20	85	82	43
Instrument variables	67	31	20		73	20	36	63	38
Hansen test ^b	0.312	0.107	0.245		0.111	0.363	0.187	0.130	0.380
AR(1)	0.082	0	0		0	0	0	0.001	0.007
AR(2) ^c	0.149	0.275	0.654		0.675	0.519	0.346	0.475	0.216

Notes. Standard errors in parentheses. ^a 2005–2022. ^b p -value. null: instruments exogenous. ^c p -value. null: no second-order autocorrelation. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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